

No. of Functional Features Included in the Solution

Date	06 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

Metric	Count / Value
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Total Features Planned	8
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S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	HTML/CSS interface allowing users to input the 4 socio-economic metrics.	Frontend (UI)	Done	High
2	Input Validation	Ensures entered values are within logical bounds (e.g., GNI > 0) before processing.	Frontend / Backend	Done	Medium
3	Data Normalization	Processes and scales the raw user inputs to match the ML training data format.	Backend Logic	Done	High
4	HDI Prediction Engine	Executes the Scikit-learn Linear Regression model to calculate the final HDI score.	Machine Learning	Done	High
5	Results Dashboard	Dynamically displays the predicted HDI score and categorizes the development tier.	Frontend (UI)	Done	High
6	REST API Endpoint	Flask routing (/predict) that receives JSON data and returns predictions.	Backend (API)	Done	Medium
7	Cloud Deployment	Live hosting of the web application making it accessible via a public URL.	Infrastructure	Done	High

Feature Summary:

Total Features Implemented	7
Core / Must-Have Features	5 (Input Form, Normalization, Prediction Engine, Dashboard, API)
Additional / Nice-to-Have Features	3 (Input Validation, Cloud Deployment, Export Results)
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	User Input Form, Results Dashboard

2	Backend / Logic	3	Data Normalization, HDI Prediction Engine, Input Validation
3	Database / Storage	1	Model Serialization (Loading hdi_model.pkl)
4	API / Integration	1	Flask REST API Endpoint (/predict)
5	Security / Authentication	1	Input Data Sanitization / Type Checking